OPEN BY DESIGN

Open Science

IS, DS, AI, and SD

A concise view of principles, infrastructures, social responsibility, and Brazil's strategic agenda.

Washington Segundo

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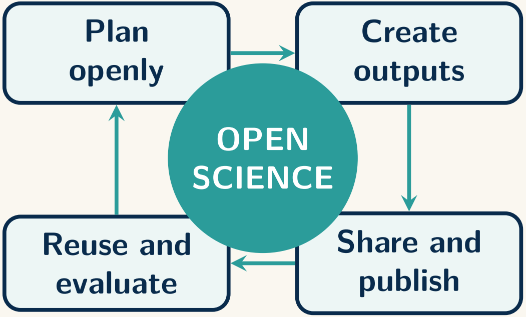
Ministry of Science, Technology and Innovation

Open science redesigns the entire research cycle



A PARADIGM SHIFT

- Openness begins at planning, not after publication.
- Publications, data, code, methods, and reviews become reusable outputs.
- Governance, preservation, and participation are part of scientific quality.

Core idea: knowledge should be accessible, auditable, interoperable, and useful to society.

Information Science is the foundation of reliable AI

Research objects → **Curation and metadata** → **Analysis and models** → **Responsible AI** → **Public value**

Research objects	Curation and metadata	Analysis and models	Responsible AI	Public value
Data, publications, code	Context and provenance	Patterns and evidence	Grounded automation	Decisions and services

Information Science

Organizes, curates, represents, and mediates knowledge

Data Science

Analyzes patterns, models phenomena, and extracts value

Artificial Intelligence

Automates cognition, prediction, and language interaction

A long evolution led from private exchange to open ecosystems

1600s	1665	1990s	2002–03	2021
Letters	**Journals**	**Web**	**Open access**	**Open science**
Private scholarly exchange	Public record and priority	Digital dis-semination	BBB declarations	UNESCO framework

Digital networks separated the traditional functions of journals: record, certification, dissemination, and preservation.

Ibict has shaped scientific information

1954

Founded as Brazil's national institute dedicated to bibliography and scientific documentation

Today
A research unit of the Ministry of Science, Technology and Innovation - MCTI

A LONG-STANDING PUBLIC MISSION

- Organize, preserve, and disseminate Brazil's scientific and technological information.
- Build national services, standards, and interoperable information infrastructures.
- Connect Brazilian research with regional and international knowledge networks.

A STRONG TRADITION OF OPENNESS Ibict has played a sustained leadership role in **Open Access** and **Open Science**, advancing repositories, aggregators, research data, digital preservation, persistent identifiers, and public-interest governance.

The ecosystem is broader than open access

Open outputs
Publications, research data, software, methods, and educational resources.

Open processes
Preprints, transparent peer review, responsible assessment, and reproducible workflows.

Open infrastructures
Repositories, identifiers, metadata standards, preservation, and research information systems.

Open participation
Citizen science, knowledge dialogue, inclusion, and public engagement.

Infrastructure connects every pillar and keeps openness sustainable.

Four principles balance technical value and social responsibility

FAIR
Make data findable, accessible, interoperable, and reusable for people and machines.

CARE
Protect collective benefit, authority, responsibility, and ethics in data governance.

TRUST
Build transparent, responsible, user-focused, sustainable, and technically sound repositories.

DEIA
Embed diversity, equity, inclusion, and accessibility in research systems.

Rule of thumb: as open as possible, as closed as necessary.

Communication and assessment are moving toward transparency

PUBLISH – REVIEW – CURATE

1. **Publish early** →
2. **Review openly** →
3. **Curate quality**

Preprints accelerate circulation; open reviews expose reasoning; curation adds context and certification.

RESPONSIBLE ASSESSMENT

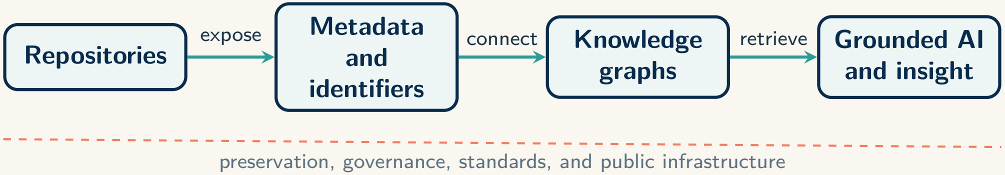
- Judge the work, not the journal.
- Value data, software, methods, review, and social impact.
- Combine expert judgment with contextual metrics.

Open access models distribute cost and power

Route	How it opens access	Main strength	Main risk
Green	Deposit in a repository	Uses public infrastructure and can avoid publication fees	Embargoes or restrictive publisher policies
Gold	Immediate access on the publisher platform	The version of record is openly available	APCs can shift exclusion from readers to authors
Diamond	No fees for readers or authors	Aligns publishing with the public interest	Requires durable institutional funding and shared governance

Publicly funded research creates maximum public value when access, publishing, and governance remain equitable.

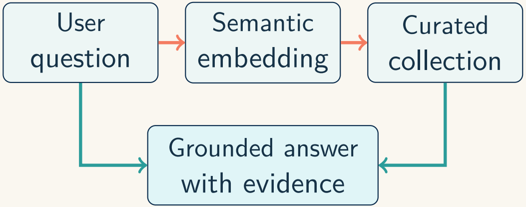
Trustworthy AI begins with trustworthy research infrastructure



Preservation and interoperability turn stored objects into reusable evidence.

Semantic systems connect meaning, not only keywords

A GROUNDED RETRIEVAL FLOW



WHAT MAKES IT WORK

- Controlled vocabularies and ontologies reduce ambiguity.
- Multilingual embeddings retrieve concepts across languages.
- RAG grounds generated answers in real documents.
- Rich metadata makes sources traceable and reusable.

Applied AI creates value when the information base is ready

Application	AI capability	Public-interest value
ServicesPI	Semantic search + NLP	Natural-language discovery across complex patent collections
Patente.IA	Text + image similarity	Broader detection of related inventions and prior art
RDAPP	NLP + semantic retrieval	Structured access and analysis across public-policy documents
SAVITS	BI + ML + LLM	Monitoring, prediction, and explainable impact assessment

The recurring architecture is simple: **infrastructure** → **structured data** → **AI application**.

Openness must also deliver informational justice

THREE DIMENSIONS

- **Distribution:** equitable access to information and infrastructure.
- **Recognition:** respect for diverse people, knowledge, and epistemologies.
- **Participation:** meaningful inclusion in producing and using knowledge.

ETHICAL SAFEGUARDS

- Consent and risk assessment
- Privacy and controlled access
- Collective and Indigenous data rights
- Protection from harmful or dual use

Open infrastructures advance a focused set of SDGs

SDG 4 – Quality Education
Open scientific knowledge strengthens higher education, research, and information literacy.

SDG 9 – Innovation and Infrastructure
Robust, interoperable digital systems provide the backbone for scientific innovation.

SDG 10 – Reduced Inequalities
Open access reduces institutional, regional, and economic asymmetries in knowledge.

SDG 16 – Strong Institutions
Transparent information systems support accountability, ethical governance, and evidence-based decisions.

SDG 17 – Partnerships
Shared standards and interoperable networks enable national and international cooperation.

Cross-cutting effect: preservation and responsible reuse also support sustainable cities, institutions, and consumption of digital resources.

Libraries are becoming critical knowledge infrastructure

AN INSTITUTIONAL EVOLUTION

1 **Collections**
Acquire and provide access

2 **Digital services**
Connect repositories and systems

3 **Infrastructure**
Steward data and long-term access

THE PROFESSIONAL ROLE EXPANDS

- Data stewardship and metadata enrichment
- Digital preservation and authenticity
- Licensing, privacy, and ethical guidance
- Persistent identifiers and interoperability
- Digital literacy and public mediation
- Coordination across research, IT, government, and society

Implementation depends on human mediation

INCLUSIVE DESIGN Build for diverse communities

- Accessible interfaces
- Multilingual discovery
- Disciplinary flexibility
- Context-sensitive workflows

GOVERNANCE Define responsibility

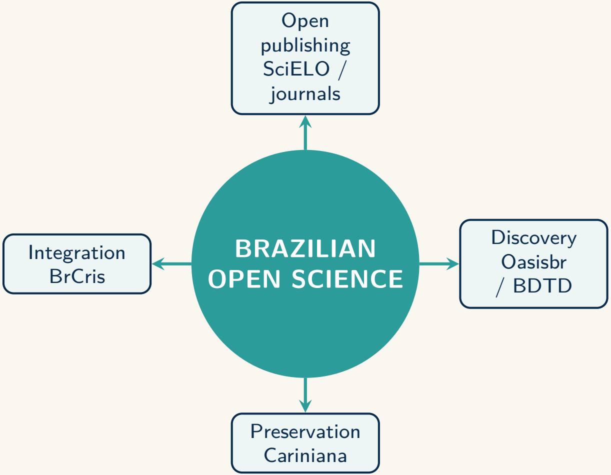
- Data control and privacy
- Curation criteria
- Ethical safeguards
- Sustainable ownership

DIGITAL LITERACY Enable critical use

- Practical training
- Limits and uncertainty
- Reproducible practices
- Responsible AI use

Multidisciplinary support teams translate Open Science principles into everyday research practice.

Brazil already has the foundations of a national ecosystem



STRATEGIC ASSETS

- SciELO and open journals
- BDTD and Oasisbr
- BrCris and research integration
- Cariniana and digital preservation
- Data, citizen science, and software networks

The next step is coordinated governance, stable funding, and stronger interoperability.

Ibict connects the national ecosystem in layers

COLLECT AND PUBLISH

- Institutional repositories and digital libraries
- Brazilian Digital Library of Theses and Dissertations
- Journal and publication support services

AGGREGATE AND CONNECT

- Oasisbr for unified discovery
- National repository networks and shared standards
- International exchange through open protocols

ANALYZE AND GOVERN

- BrCris for research information and policy intelligence
- Laguna for semantic integration and large-scale analysis

PRESERVE AND IDENTIFY

- Cariniana for distributed digital preservation
- dARK for persistent, traceable identifiers

Together, these layers preserve institutional autonomy while enabling national-scale discovery and reuse.

Scientific infrastructure enables sustainable development

Integrated data → Accessible knowledge → Evidence-based innovation → Sustainable public benefit

Integrated data
Open, governed, interoperable

Accessible knowledge
Structured and discoverable

Evidence-based innovation
Research translated into action

Public benefit
More resilient and equitable systems

Three complementary infrastructures strengthen Brazilian research

BRCRIS Connects the research system

- Researchers and institutions
- Projects and funding
- Outputs and infrastructures

Enables visibility, governance, assessment, and evidence-based policy.

DARK NETWORK Keeps resources identifiable

- Persistent ARK identifiers
- Traceability and citation
- Long-term access

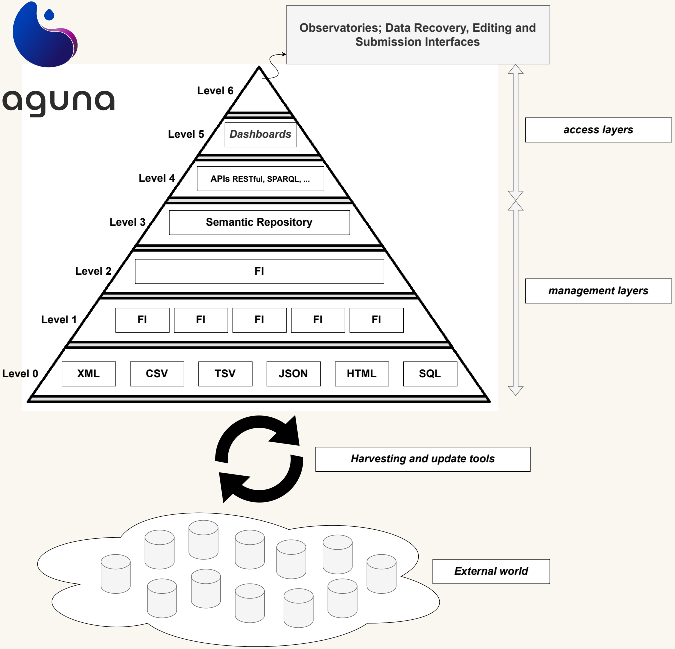
Protects reproducibility and institutional memory from broken links.

PROJECT LAGUNA Integrates heterogeneous data

- Semantic enrichment
- Interoperability
- Large-scale discovery

Transforms isolated sources into a shared analytical environment.

Laguna



Observatories; Data Recovery, Editing and Submission Interfaces
Level 6
Level 5
Dashboards
Level 4
APIs RESTful, SPARQL, ...
Level 3
Semantic Repository
Level 2
FI
Level 1
FI
FI
FI
FI
FI
Level 0
XML
CSV
TSV
JSON
HTML
SQL
access layers
management layers
Harvesting and update tools
External world

BrCris creates a national view of Brazilian research

BRAZILIAN CURRENT RESEARCH INFORMATION SYSTEM

- Developed and led by **Ibict**; officially established in 2014.
- Integrates, retrieves, certifies, and visualizes information about Brazil's research ecosystem.
- Organizes fragmented records into a unified national model aligned with international standards.
- Provides free access for researchers, managers, funders, journalists, and citizens.

15.9M

publications indexed

Platform count displayed in July 2026

A public gateway to understand who produces knowledge, where, with whom, and with what results.

BrCris connects actors, outputs, and infrastructures

RESEARCH ACTORS

- Researchers
- Institutions
- Research groups
- Graduate programs
- Funders and projects

RESEARCH OUTPUTS

- Publications
- Theses and dissertations
- Scientific datasets
- Patents
- Software

INTEGRATED SOURCES

- Lattes and CNPq directories
- Oasisbr and Capes/Sucupira
- OpenAlex and OpenAIRE
- DOAJ, INPI, and Espacenet
- NDLTD, ROR, and Wikidata

Source: official BrCris About page and FAQ, accessed 28 Aug. 2026.

Integration turns records into policy intelligence

INFORMATION PIPELINE

1. Collect records from national and international sources.
2. Standardize, normalize, and deduplicate the metadata.
3. Connect people, organizations, projects, and results.
4. Provide search, individual profiles, and indicator dashboards.

DECISIONS IT CAN SUPPORT

- Map collaboration networks and scientific communities.
- Compare institutional and thematic research profiles.
- Identify strengths, gaps, and emerging trends.
- Improve transparency, visibility, and research assessment.
- Ground science, technology, and innovation policies in evidence.

Source: official BrCris About page, Dashboards, and FAQ, accessed 28 Aug. 2026.

Search in the Brazilian Scientific Research Information Ecosystem (BrCris)

Publications | Enter at least 3 characters and search among 15,910,175 Publications | Search

BrCris

The Brazilian Scientific Research Information Ecosystem, BrCris, is an aggregator platform that allows retrieving, certifying and visualizing data and

Oasisbr connects more than 6.5 million open digital objects

A NATIONAL AND INTERNATIONAL GATEWAY

6.5M+

articles, theses, dissertations, books, book chapters, and research datasets

~1.2M

documents aggregated through BDTD

~2,000

information sources — repositories and scientific journals across Brazil

BDTD gives visibility to graduate research

A NATIONAL SERVICE SINCE 2002

- Conceived, developed, and maintained by Ibict.
- Integrates theses and dissertations from Brazilian teaching and research institutions.
- Also includes records of works defended by Brazilians at institutions abroad.
- Offers free national discovery; full texts remain hosted by participating institutions.

PUBLIC VALUE

- Visibility for postgraduate research
- Shared metadata standards
- Institutional interoperability
- Access to specialized knowledge
- Accountability for public investment

Oasisbr, BDTD, and BrCris play distinct roles

Service	Primary question	Scope	Strategic role
Oasisbr	What open research outputs can I discover?	Publications, data, books, conference works, theses, and more	Multidisciplinary access and national/international aggregation
BDTD	What graduate research has been produced?	Brazilian theses and dissertations, including records from abroad	Visibility and governance of postgraduate knowledge
BrCris	How is the research ecosystem connected?	Actors, organizations, projects, programs, outputs, patents, and software	Research intelligence, indicators, governance, and policy support

Together they connect **discovery**, **specialized graduate knowledge**, and **system-level intelligence**.

Sources: official Oasisbr, BDTD, and BrCris portals, accessed 28 Aug. 2026.

dARK turns persistence into regional sovereignty

WHAT IT ADDS TO ARK

- **Decentralization:** participating institutions share operational responsibility.
- **Traceability:** identifier events remain auditable over time.
- **Resilience:** distributed governance reduces single points of failure.
- **Sovereignty:** regional institutions define policies and priorities.

HOW IT WORKS WITH OASISBR

1. Repositories expose qualified metadata through open protocols.
2. Oasisbr aggregates and validates the records.
3. dARK assigns persistent identifiers to eligible objects.
4. Users discover content centrally while access remains with the source institution.

dARK

Brazil needs a national implementation plan

POLICY ARCHITECTURE

- A national policy with clear public-interest goals
- Defined roles for funders, institutions, and scientific societies
- Shared standards with room for regional and disciplinary adaptation
- Stable funding for open infrastructures and professional teams

DELIVERY MECHANISMS

- Phased mandates linked to practical support
- Incentives aligned with responsible assessment
- Training, monitoring, and transparent indicators
- Continuous consultation with researchers and society

The objective is gradual, consistent change that increases the public return on research investment.

The agenda now shifts from isolated projects to durable policy

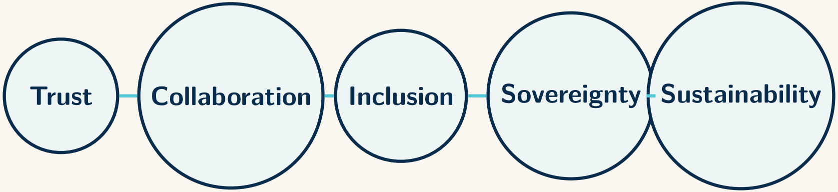
SYSTEM PRIORITIES

- Sustain public-interest infrastructures.
- Integrate platforms, standards, and preservation.
- Train researchers, librarians, data stewards, and developers.
- Reform incentives and research assessment.
- Govern quality, privacy, bias, and accountability.

RESEARCHER ACTIONS

- Publish through open routes.
- Share documented data and code.
- Use persistent identifiers and open licenses.
- Join transparent review and citizen science.
- Plan semantics, governance, and preservation from the start.

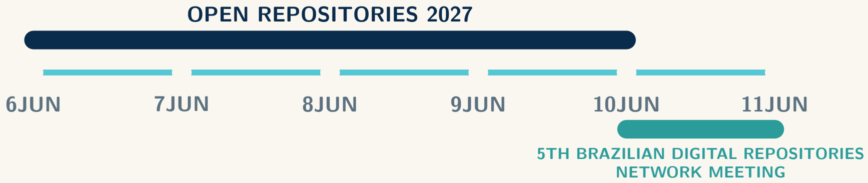
Open science is a strategic investment



Infrastructure + public policy + scientific practice + social movement

Together, they make research more reliable, reusable, inclusive, and aligned with the public interest.

Salvador/Brazil connects repository communities in 2027



6–10 JUNE 2027

Open Repositories 2027

Salvador, Bahia — global exchange comes to South America's repository community.

10–11 JUNE 2027

Brazilian Repository Network

The fifth national meeting advances cooperation and shared repository practices.

One city. Two connected events. Six days of repository collaboration.

Thank you!

washingtonsegundo@ibict.br

Aguyjevete
Guarani

Sulpayki
Quechua

Asante
Swahili / Bantu

Ngiyabonga
Zulu / Bantu

Ẹ ṣé
Yoruba

Na gode
Hausa

धन्यवाद
Hindi

ধন্যবাদ
Bengali

நன்றி
Tamil

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Amharic

Terima kasih
Indonesian

谢谢
Chinese

Obrigado
Portuguese